

Preliminary Design Proposal: Automated Stage-Top Incubator for Live-Cell Imaging

1. System Overview & Purpose

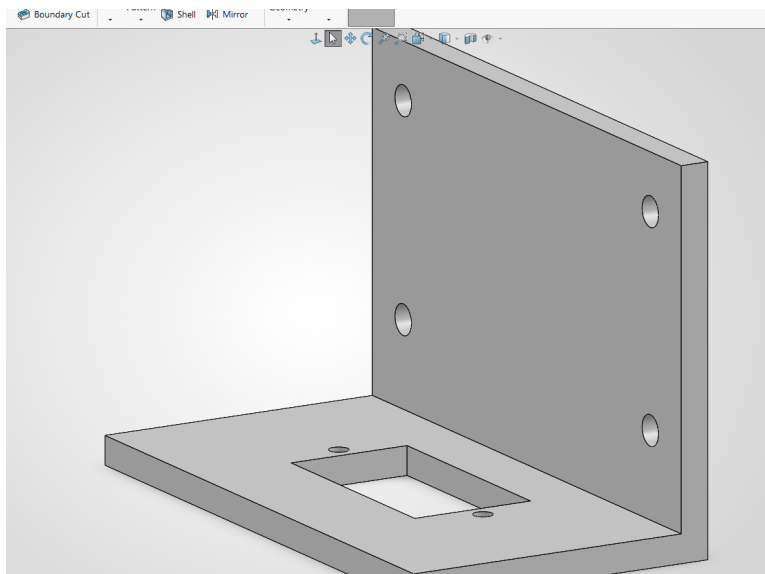
This document details the mechanical specifications and rapid-prototyping pipeline for an automated stage-top live-cell incubator. The purpose of this device is to maintain strict physiological conditions (37°C, 5% CO₂, high humidity) for mammalian cell cultures directly on the stage of an inverted microscope, enabling long-term, time-lapse imaging without requiring a massive, expensive full-microscope enclosure.

The core feature of this design is an automated rack-and-pinion carriage mechanism that allows a standard 96-well microplate to be programmatically ejected and retracted. This enables seamless robotic liquid handling or manual media changes without disrupting the internal thermal environment or optical alignment.

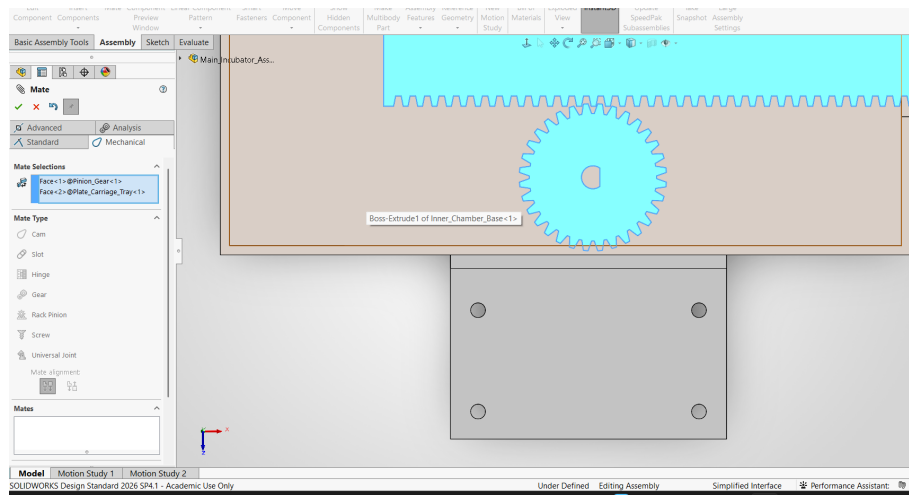
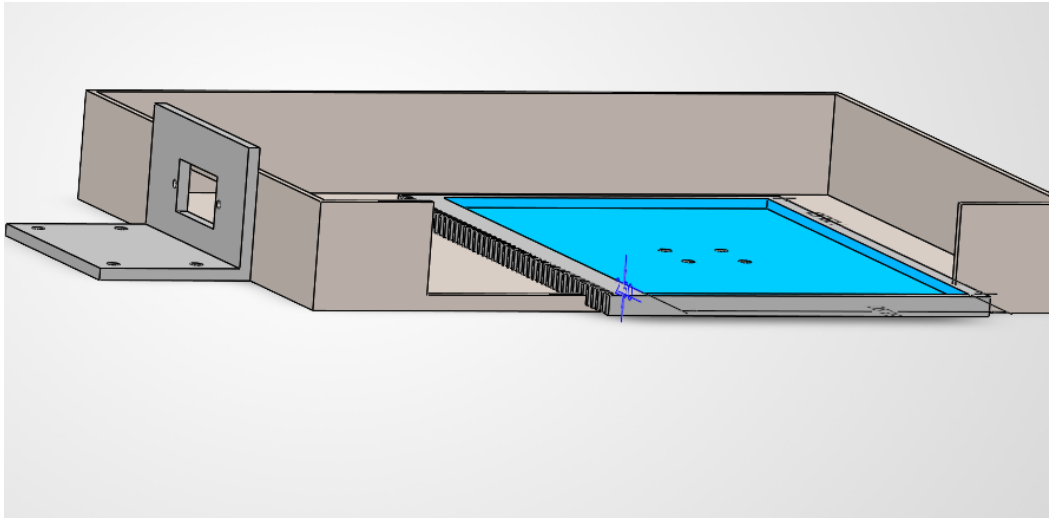
2. Mechanical Design & Assembly

The structural assembly is engineered for modularity. It consists of five primary components:

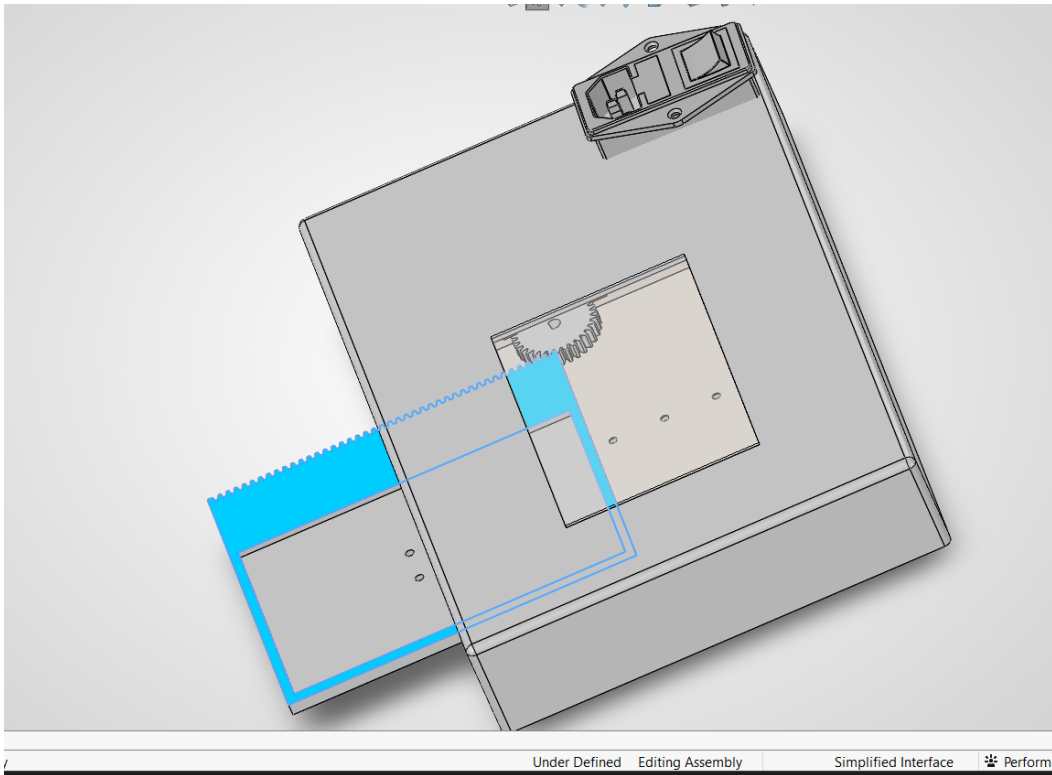
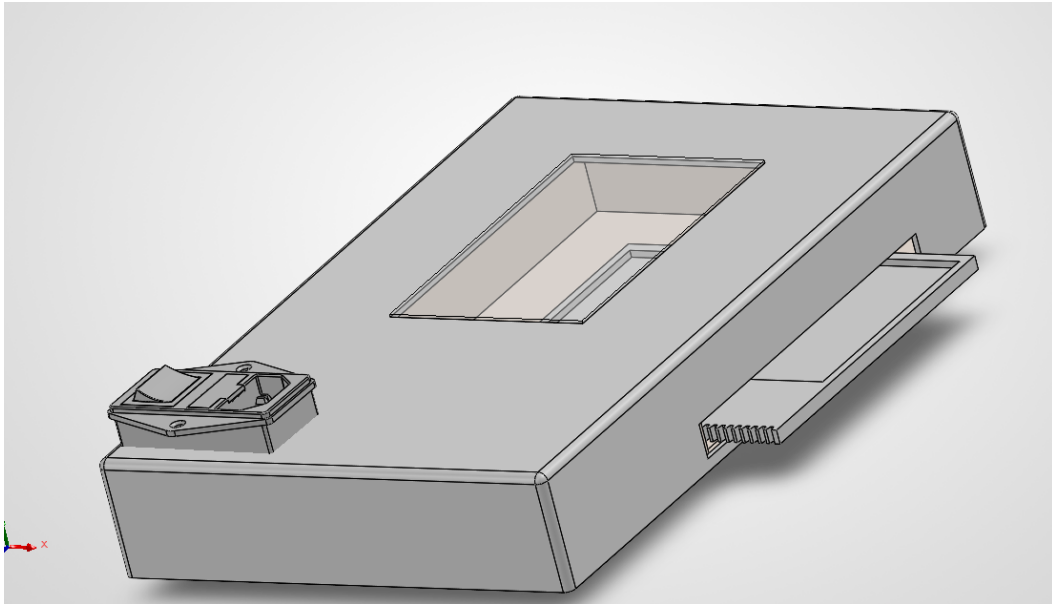
- **Motor Bracket:** An L-shaped mounting fixture featuring four through-holes to secure the actuator away from the heated biological zone.



- **Inner Chamber Base & Carriage Tray:** The primary containment floor houses the sliding carriage tray. The tray accommodates a standard 96-well microplate and features an integrated linear rack gear along the longitudinal edge for automated ejection and retraction.
- **Drive Mechanism:** A rotary pinion gear meshes with the carriage tray's rack gear, translating rotary motor output into precise linear translation of the cell cultures.



- Outer Shell & Internal Cavity:** A unibody enclosure featuring an offset rectangular viewing window (aligned for inverted microscopy), a lateral ejection port, and an IEC power switch module. The solid, opaque section of the shell forms an internal “side-car” cavity. This isolates the mechanical drive and potential control electronics from the humidified incubation zone.



3. Manufacturing & Fabrication Plan

To ensure this hardware can be deployed rapidly on-site, the entire structural chassis is optimized for Fused Deposition Modeling (FDM) 3D printing. This eliminates lead times associated with CNC machining and allows for overnight geometric iterations.

- **Structural Components (Shell, Chamber, Bracket):** Fabricated on-site using PETG or ABS filament. These thermoplastics provide the necessary thermal stability to prevent warping under sustained 37°C incubation conditions.
- **Drive Components (Gears):** Printed in Nylon or POM (Acetal) to ensure low friction and high wear resistance during continuous automated oscillations.
- **Optical Window:** A 1mm insert of clear polycarbonate or standard silicate glass will be manually cut and sealed. FDM transparent filaments will not be used, as they lack the optical clarity required for high-resolution inverted microscopy.

4. Two-Month On-Site Hardware Implementation Timeline

The following 8-week timeline is designed to aggressively finalize and assemble the physical hardware. Completing the mechanical build within this two-month window ensures that subsequent time is entirely dedicated to software tuning, PCB reprogramming, and rigorous validation via live-cell assays.

- **Weeks 1–2: Dimensional Verification & CAD Finalization**
Measure the specific inverted microscope translation stage. Modify the inner chamber base CAD to include exact mounting flanges or dovetail grooves. Finalize the internal cavity dimensions based on the chosen microcontroller footprint.
- **Weeks 3–4: Rapid Prototyping & Mechanical Assembly**
FDM print the inner chamber, motor bracket, and carriage tray in PETG. Print the rack and pinion gears in Nylon. Assemble the drive mechanism and physically verify smooth, jam-free linear translation of the tray.
- **Weeks 5–6: Outer Shell Fabrication & Subsystem Integration**
Print the unibody outer shell. Install the motor, limit switches, IEC power socket, and physically mount the custom PCB inside the side-car cavity. Tap pneumatic ports for CO₂/N₂ gas lines and install the glass optical window.
- **Weeks 7–8: Environmental Calibration & System Handover**
Seal the chamber and conduct initial dry runs. Validate that the internal chamber reaches and holds 37°C without deforming the plastics. Hand over the completed physical hardware for the secondary project phase: PCB firmware tuning, UI software development, and live-cell testing.

5. Open Integration Dependencies

Before Week 1 commences, the following three parameters must be finalized with the laboratory:

1. **Microscope Placement:** Does the target microscope require the incubator to be hard-mounted via screws/flanges, or will it rest passively on the translation stage?
2. **Gas Routing (CO₂/N₂):** Will the system tap into an existing external pneumatic mixer, requiring only push-to-connect tube fittings on the shell?

3. **Control Electronics:** Should the custom mechatronic PCB be housed internally within the shell's side-car cavity, or routed to an external desktop control box?